

Partial Differential Equation Agents (PDE-Agents): A Large Language Model-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning

Highlights

- Multi-agent large language model system automates finite element simulations
- Graph-based retrieval-augmented generation provides grounded reasoning
- Knowledge graph smart integration achieves 100% on 50 benchmark tasks
- Novel-material experiment proves knowledge graph value for unseen data
- Verified $\mathcal{O}(h^2)$ convergence on two non-trivial benchmarks